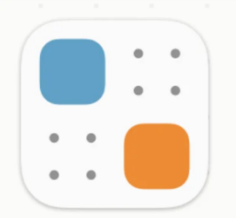


Patch Pure Data

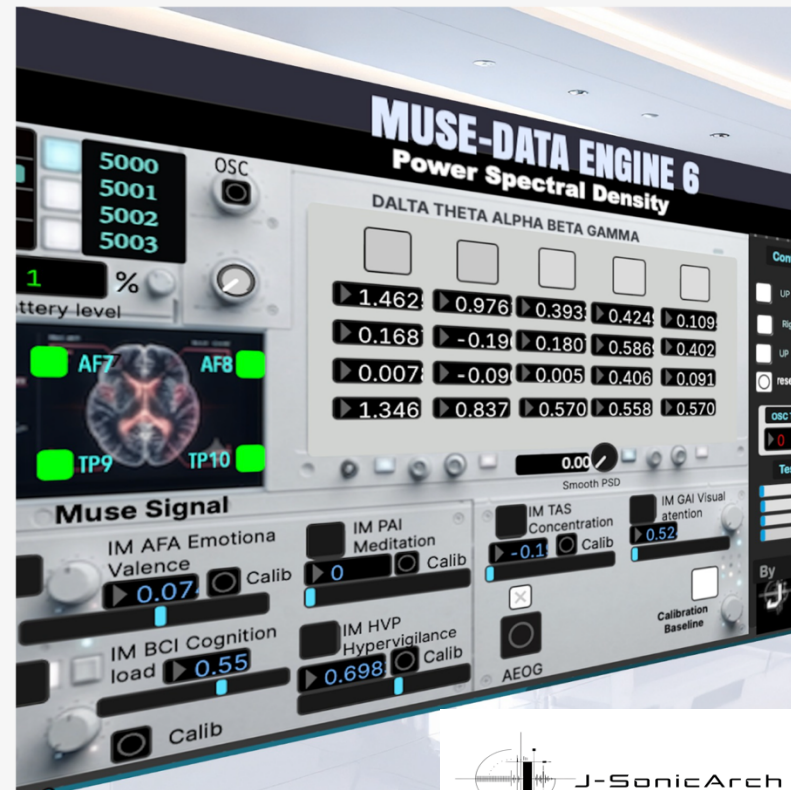


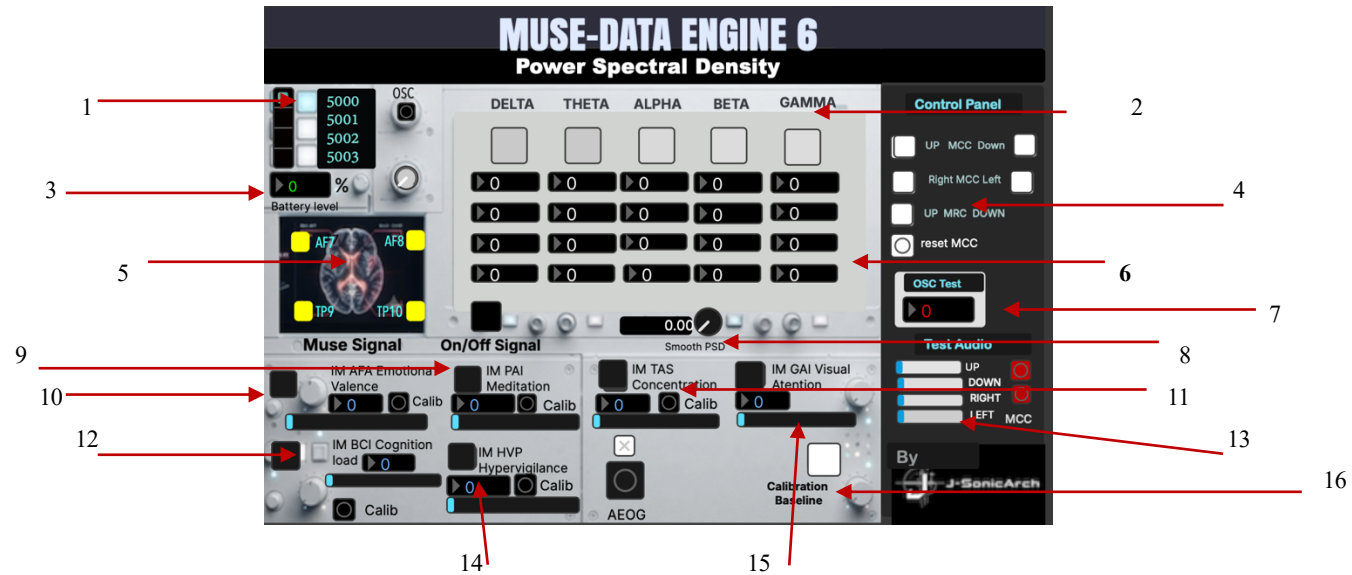
PLUG DATA

MUSE DATA ENGINE V 6.0 MODULE

Diseño y desarrollo de sistemas MMHCI híbridos con bioseñales y un DMI de smartphones, para obras bio-interactivas mixtas y performáticas

Pure Data Patches for the PhD Thesis: Juan Pablo posada Alvarez





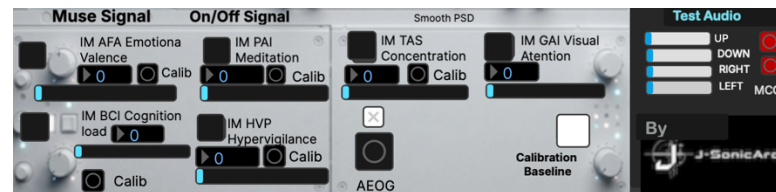
A general guide to the functions of the Engine module is provided below. This version, featuring a simpler visual interface, allows for the optimization of the computer's CPU and memory resources.

1. **OSC port:** This module allows the selection of the data reception port using the standard **OSC** protocol used for real-time signal transmission. Available ports (5000, 5001, 5002, and 5003) offer flexibility in configuring data streams, making it easy to integrate with multiple external devices or applications
2. **PSD Stock Section** Single band sec-on
On/Off (Brain Rhythm Bands): This button, integrated into each brain rhythm band (alpha, delta, beta, gamma), controls the activation or deactivation of data reception using the OSC protocol from the Mind Monitor application. When disabled, system

and PMSon interactions are temporarily suspended. This option allows the user to selectively manage the display of bands of interest.

3. Device Battery Level
4. **IMmove-Data control button:** Each button enables MIDI output for continuous and rapid head movements along the X and Y axes.
5. **Signal quality information on all electrodes**
Green: Receiving good signal in the electrode.
Yellow: Receiving medium quality signal in the electrode
Red: receives a signal with a lot of noise
6. Individual PSD values for each (Brain Rhythm Bands)
7. **OSC send test Data:** When the values are changed, they are sent via the OSC protocol, serving as a tool for testing data transmission.
8. **Window Adjustment (Moving Average Filter) PSD Values:** Adjusting the filter window, specifically the Moving Average filter, allows you to modify parameters that directly influence the flow of PSD values. S should adjust this value only in cases where the values have rapid and abrupt variations, in most cases a value of 4 adjusts the signal values appropriately.

EEG Interaction Section



This interface presents the approximate values of each IM designed for the EEG interaction module using a set of visual *sliders*. Each descriptor is associated with a specific neurophysiological parameter, as detailed below

9. **Meditation Section:** In this section, IM PAI interaction is enabled or disabled. The button for calibrating the baseline or basal state is also located here. The slider indicator reflects changes in the performer's states of calm or meditation. When the *slider* is positioned to the left, states of low meditation are detected, tending to a state of alertness or active attention. On the contrary, a shift to the right indicates high values associated with states of meditative calm.
10. **Emotional Valence Section:** In this section, IM AFA interaction is enabled or disabled. The button for calibrating the baseline or basal state is also located here. Emotional *Valence*: This slider provides a visual guide to the performer's estimated emotional valence (positive or negative). A shift to the left indicates values associated with a negative valence, while a shift to the right indicates a positive valence.
11. **TAS Attention Section:** In this section, IM TAS interaction is enabled or disabled. The button for calibrating the baseline or basal state is also located here. The slider indicate reflects changes in the performer's states attention. When the *slider* is positioned to the left, states of low attention are detected, tending to a state of distraction. On the contrary, a shift to the right indicates high values associated with states of Attention or concentration.
12. **BCI cognitive load Section:** In this section, IM BCI interaction is enabled or disabled. The button for calibrating the baseline or basal state is also located here. The slider indicate reflects changes in the performer's cognitive load in performance. When the *slider* is positioned to the left, states of low cognitive load are detected. On the contrary, a shift to the right indicates high values associated with high states of cognitive load.
13. **IMmov test:** This section is linked to MUSE IMmove-Data motion. Here, you can test the transmission of MIDI data for head movement.
14. **HVP hypervigilance section:** In this section, IM HVP interaction is enabled or disabled. The button for calibrating the baseline or basal state is also located here. The slider reflects changes in the interpreter's hypervigilance states. When positioned to the left, it detects states of low vigilance, characterized by visual attention and rapid, synchronized blinking; conversely, a shift to the right indicates high values associated with hypervigilance.

15. **GAI visual attention:** In this section, IM GAI interaction is enabled or disabled. The button for calibrating the baseline or basal state is also located here. The slider reflects changes in the interpreter's Visual Attention states. When positioned to the left, it detects states of low visual concentration, characterized by not fixing the gaze on a single point. A shift to the right indicates high values associated with a fixed gaze.
16. **Baseline adjustment:** This button allows you to adjust the baseline values for all interactive cognitive tasks (IM). This option is useful when you need to adjust all user neuromarkers simultaneously.

Outlet 1: Connection to the MUSEDDataIMmov module. If this connection is not established, the functions associated with:

- Motion detection.
- PPG and fNIRS sensors